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PROJECT
BRIDGE DESIGN AND FABRICATION

1. Project Description

The purpose of this project is to analyze the loading conditions of a gantry bridge and redesign the bridge such that it can bear the loading requirements stipulated below. As you have gained sufficient knowledge about Stress Analysis from this class, it is time to put your knowledge to the test. In this project, you are provided a bridge deck (1000 X 100 X 3 mm) made from PMMA, aka Acrylic. You are to build a complete bridge that satisfies the following conditions:

1. The bridge shall span a chasm that measures 900 mm from one edge to the other.
2. The bridge has a gap in the middle. This gap is to allow a tool mounted to a vehicle travelling through the bridge to access the area underneath the bridge. This gap should not be obstructed under any circumstances. If this gap is obstructed, your design will be rejected.
3. The bridge shall be made entirely of Acrylic and held together by glue. No mechanical fastener shall be allowed.
4. The bridge shall have guard rails and allow for 1 way traffic. The traffic will be for 2 PAScars (link). The guard rails should be implemented for the 900 mm length only (not including the section where the bride is resting on the table).
5. The cross-section area of the bridge including guard rails is required to be symmetric, as shown in below figure.
6. The combined payload of these PAScars shall be 8 kg maximum not including the weight of two PAScars. The bridge's allowable stress is recommended to have a tensile safety factor of 1.5 when under this load.
7. The bridge shall not be damaged or disintegrated in any way as the PAScars travel through it.
8. The maximum deflection point on the bridge shall not be more than 7 mm as the PAScars travel through it.
9. The bridge shall connect seamlessly to the 1.2 meters PASCO dynamics track (link) at both ends. "Seamlessly" means the PAScars should be able to travel to and from the dynamic tracks without any hindrance. The dynamic tracks will be laid on top of the table. You are not allowed to modify the dynamic tracks in any way.
10. The bridge shall only be fix-supported at 1 end and simple-supported at the other end (which means the bridge is just resting on top of the table on that end). The fix-support shall be done by 2 hand-clamps at the first 50 mm length of the bridge (without any guard rail). The simple support will be facilitated by the surface of the table where the bridge terminal point is resting. No human contact shall be permitted to provide support for the bridge.
11. The bridge shall bear permanent markings to identify which team it belongs to. Failure to do so could lead to your bridge being lost and you will receive no credit for the demonstration.
12. No Computer Simulation or Analysis result will be recognized in this project. If you submit an analysis done by a CAD/CAE or Simulation package (e.g., Finite Element Analysis by Solidworks), you will receive a zero for the entire result.

The physical characteristics of Acrylic had been found by examining the results of a Tensile Test on an acrylic specimen as follows.

- Young's Modulus of the material is 1.1 GPa
- Yield Strength of 24.4 MPa

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- Shear Yield Strength, 14.1 MPa
- Density, 0.043(lb/in³)- 1.18 g/cm³)
- Passion's ration – 0.37 The material will fracture after it deforms plastically.

To strengthen the bridge deck, you may want to **furnish extra parts to modify its cross-sectional area**. For this purpose, you are authorized to use the Laser Cutter in our Advanced Manufacturing Workshop to cut the appropriate parts according to your design. Please note the following restrictions:

1. The only stock available for this project is Acrylic with 3 mm thickness.
2. Each team is allocated up to 0.25 m2 (or 2500 cm2) of material, not counting the bridge deck provided initially. Each extra means a deduction on your total grade for the project as below Table 1.

Table 1 Extra material vs. deduction

No.	Extra material (cm2)*	Deduction of total grades
1	100	1%
2	200	2%
3	300	3%
4	400	4%
5	500	5%

*Area in 2-Dimentions

3. You can submit a fabrication request by sending an email to the lecturer (Mr. Thanh Tran, thanh.tran37@rmit.edu.vn) to declare your intentions. The email should have the subject **"MIET2499: Fabrication from team XX"** (Where XX is the name or number of your team). The request should be a zipped folder, containing a pdf file containing the Bill of Materials which notes the part names, description, surface area, and desired quantities, together with an isometric view of the bridge. The part designs shall be captured in a .dxf file and Solidworks Part files. Only Solidworks files will be recognized. The drawing unit must be millimeters. If possible, please include an assembly file with an exploded view of the bridge design for better examination and feedback.
4. Your team is responsible for assembling the parts into the bridge deck. You are free to use any glue you can procure.
5. The deadline for submitting a fabrication request is **Saturday, September 14th, by 11:59 Pm (I suggest that you may submit earlier, in which case you will receive the parts earlier)**.

The bridge should be ready for testing with actual loads on **Wednesday, September 20th, during your usual Tutorial hours**. The tests shall be conducted in the same classroom as your tutorial session. The grade for this test is worth 10% of your total grade.

The following items are available for you to check out from the Robotics. The below items will show you in the tutorial session of the Week 8.

A. Electronic Digital Caliper.
B. PAScars.
C. PASCO Hook Masses.
D. PASCO 250g Bar Masses.
E. PASCO Aluminum Tracks.

If you need any additional items, please contact our technician, Mr. Tai (tai.vo3@rmit.edu.vn).

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2. Project requirements

1. From the details provided in the Project Description, perform the calculations and analysis necessary to produce a new design for the bridge deck. The drawings, calculations, analysis, and descriptions shall be captured using the **Project Report Template** (available on Canvas). **The submission deadline for the Design Report is Saturday, September 25th, 2024, by 11:30 AM**.
2. All project team members are required to attend the **Demonstration session on Friday, September 20th, during your usual Tutorial hours**, during your usual Tutorial hours, in the Robotics Lab, room 8.1.002. Absence from the session means a zero for the testing portion of that student.
3. The Design Report should have a high level of detail with S.M.A.R.T. (Specific, Measurable, Achievable, Relevant, Time-based) descriptions.
4. All sections within the report must be fulfilled. In case it cannot be filled, note the reason why. Make sure you discuss the reason with the Lecturer beforehand, otherwise you may lose points to that portion.
5. Your fabrication files (.dxf format) can be submitted at any time between Week 8 and Saturday, Sept 14th, by 11:59 Pm. If there is any problem with the design, you will need to modify your design and resubmit it within 2 days. Fabrication is carried out on a first-come-first-serve basis so you may need to wait a long time to receive your parts if you submit your design late.
6. All submitted designs, work, and report must be original. **If plagiarism is detected, penalties and disciplinary actions will apply per RMIT Code of Conduct**

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